

User: Hi GEPETTO. I need to prepare a Domain-Driven Design proposal for a beauty salon business owned by Isabella. Help me understand the domain before I write the proposal.

GEPETTO: Great. Let us start with domain exploration. A beauty salon usually coordinates customer bookings, employees, treatment rooms or chairs, service durations, product usage, payments, and customer communication. The main business goal is to deliver services efficiently while keeping staff calendars full and customers satisfied.

User: What are the likely core business processes in such a salon?

GEPETTO: Typical core processes are: customer registration, appointment booking, service selection, staff assignment, check-in, service delivery, payment collection, rebooking, cancellation and rescheduling, and follow-up communication such as reminders or promotions.

User: What services should I assume Isabella offers?

GEPETTO: A realistic small-to-medium salon can offer haircuts, coloring, blow-dry, manicure, pedicure, makeup, skincare, waxing, and package services. Some services may require only one staff member, while others may require a specific skill set, fixed duration, special equipment, or consumable products.

User: Without software, what pain points would the salon probably face?

GEPETTO: Common pain points are double bookings, missed appointments, lost customer history, difficulty tracking employee schedules, unclear payment records, weak reporting, forgotten follow-ups, and slow communication when appointments change. Manual notebooks and phone-based booking usually create errors and waste time.

User: What should the system goals be?

GEPETTO: The system should centralize customer data, simplify scheduling, reduce booking conflicts, support staff coordination, record payments accurately, automate reminders, and provide reports for business decisions. It should also improve customer experience by making bookings faster and more transparent.

User: Please outline the functional areas I should include.

GEPETTO: You should include at least: customer management, appointment scheduling, staff management, services catalog management, payments and billing, notifications, and reporting. Optional extensions include inventory, loyalty programs, and online customer self-booking.

User: For customer management, what information matters most?

GEPETTO: Customer profile, phone number, email, preferred contact method, visit history, allergies or preferences if the salon stores them, notes, and loyalty or discount eligibility. In DDD terms, Customer is a key business entity because many processes depend on customer identity and history.

User: What does appointment scheduling need to handle?

GEPETTO: Appointment scheduling must manage date and time, selected services, assigned employee, estimated duration, status, price estimate, room or chair availability if relevant, cancellation or reschedule rules, and reminder timing. It should prevent overlap for the same employee and ensure only qualified staff can be assigned.

User: What about staff management?

GEPETTO: Staff management should cover employee profiles, roles, working hours, availability, skills, leave days, service eligibility, and optionally commission rules. A stylist or beautician should only be bookable for services they are trained to provide.

User: Payments and billing?

GEPPETTO: The system should record invoice or receipt number, appointment reference, payment amount, payment method, discounts, taxes if applicable, payment status, and refund information. It should support split or partial payments only if Isabella's salon needs them; otherwise keep it simple.

User: And notifications?

GEPPETTO: Notifications usually include appointment confirmations, reminders, reschedule alerts, cancellation notices, and possibly promotional messages. Internally, the staff may also receive alerts about today's schedule changes.

User: What reports would be most useful for the owner?

GEPPETTO: Daily appointments, revenue by period, revenue by service, revenue by employee, cancellation rate, busiest hours, customer return frequency, and unpaid balances if the salon allows pay-later scenarios.

User: Help me identify the main domain entities or aggregates.

GEPPETTO: A practical conceptual model would include Customer, Employee, Service, Appointment, Payment, Invoice or Receipt, Notification, and possibly StaffSchedule. Appointment is often the operational center because it connects customer demand, staff supply, services, time, and money.

User: Which business rules are important enough to mention explicitly?

GEPPETTO: Important rules include: an appointment must belong to one customer; an appointment must contain at least one service; assigned staff must be qualified for all services they perform; overlapping appointments for the same employee are not allowed; canceled appointments cannot be paid unless there is a cancellation fee rule; completed appointments may trigger billing and follow-up notifications.

User: Are there any value objects I can mention?

GEPPETTO: Yes. Money, TimeRange, ContactInfo, ServiceDuration, and maybe Address are good value objects. They describe concepts by their values and are not usually tracked by identity.

User: Can you suggest user roles and permissions?

GEPPETTO: A strong starting point is: Owner or Administrator, Receptionist, Staff Member, and Customer for online booking. The owner has full access. Receptionists manage bookings and customer records. Staff members view their schedule and update appointment progress. Customers can view and book their own appointments if a portal exists.

User: What would a high-level database structure look like?

GEPPETTO: Use core tables such as customers, employees, services, appointments, appointment_services for the many-to-many between appointments and services, payments, notifications, and employee_availability or schedules. Add audit fields like created_at and updated_at to each transactional table.

User: Is there a simple bounded-context view I can mention without going too deep?

GEPPETTO: Yes. You can separate the domain into Booking Management, Customer Management, Staff & Availability, Billing, and Communication & Reporting. This keeps the proposal aligned with DDD thinking without requiring full tactical modeling.

User: Please give me a final summary I can use in a professional proposal.

GEPPETTO: Summary: the salon domain revolves around coordinating customers, services, employees, time slots, and payments. The proposed software should reduce operational friction, improve scheduling accuracy, preserve customer history, automate routine communication, and give Isabella visibility into

revenue and staff performance. A clean conceptual model centered around Appointment, Customer, Employee, Service, and Payment will support future extensions such as online booking or loyalty features.